Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/19/2025
PI Name:	Wyrobek, Andrew Ph.D.		
Project Title:	Molecular Characterization of Choroid Plexus and Hippocampal Damage and Degenerative CNS Risks from Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	For immediate assistance please contact my administrator Caron LaMarsh clamarsh@lbl.gov, 510.486.5317		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	04/04/2014	**End Date:**	01/22/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/22/2022 per R. Elgart/HRP (Ed., 6/25/21) NOTE: End date changed to 1/22/2021 per J. Weeks/HRP (Ed., 1/28/2020)		
Key Personnel Changes/Previous PI:	February 2021: Key members are: Laurent-Philippe Albou, Chris Mungall, Ben Brown, Lawrence Berkeley National Laboratory (LBNL); Halina E. Witkowska, University of California, San Francisco; Bernard Rabin, University of Maryland, Baltimore Campus; Richard Britten, Eastern Virginia Medical School; Tore Straume, NASA Ames; Ana Mora, Robert Gunier, University of California, Berkeley.		
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Grant/Contract No.:	NNJ14HP06I
Performance Goal No.:	
Performance Goal Text:	
Task Description:	A major goal of the NASA space radiation research program is to enable human exploration of space without exceeding limits for immediate and persistent risks to the central nervous system (CNS) from space radiation. Our research plan builds on the mounting evidence of the radiation-sensitivity of the hippocampus, which is important for memory and performance, and the choroid plexus (CP), which produces cerebral spinal fluid and maintains the blood-brain barrier. Using transcriptomic, proteomic, metabolomics, and immunohistochemical analyses of CNS tissue from rodents exposed to simulated space radiation, we have identified molecular and cellular changes associated with oxidative stress, neuro-inflammation, vasculature damage, changes in the protein composition of cerebral spinal fluid, and amyloid deposits such as those found in Alzheimer disease (AD). The ongoing research investigates the low-fluence responses and post-exposure-time dependence of these deficits, to identify the molecular mechanisms of inter-individual variations in behavioral responses after radiation exposure, and to validate the predicted mechanisms using tissue of irradiated and behaviorally characterized rodents. Our specific aims are to: (1) Investigate the cellular and functional abnormalities in the hippocampus and CP of HZE (high energy)-irradiated rats that are predicted by our multi-omics findings with the emphasis on defects associated with AD. (2) Investigate hippocampal and CP radiation damage responses associated with inter-individual variation in radiation-induced neurocognitive deficits, with emphasis on tissue defects associated with AD. This research is foundational to establish quantitative mechanism-based predictors for neuro-cognitive risks in individuals exposed to high-LET (linear energy transfer) space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is designed to provide NASA with deeper mechanistic understandings to reduce uncertainties in risk estimates for behavior deficits (during mission) and individual susceptibility for future neuropathologies and neurological diseases (after mission) arising from space flight and exposures to space radiation. This project (a) employs state-of-the-art molecular profiling and tissue validation technologies to understand the mechanisms of the CNS damage response in animals exposed to low-fluence HZE particles (i.e., <1 hit per cell), (b) links the type and magnitude of molecular defects with behavioral performance, (c) validates the predicted molecular neuro-pathology in brain tissue, and (d) utilizes outbred rat strains to mimic individual variation in radiation damage responses. Our project focuses on the hippocampus (site of spatial memory and anxiety responses) and choroid plexus (CP, small cauliflower-like tissue in the ventricles adjacent to the hippocampus that produces cerebral spinal fluid (CSF) and maintains blood-brain homeostasis). This project will distinguish between two broad categories of risk: (1) mechanisms related to the average response to exposure dose, and (2) mechanisms of inter-individual variations in risks for behavioral performance during the mission, neuropathologies, and late-onset neurological diseases (i.e., Alzheimer's Disease). Our research is conducted in outbred strains of rats to capture the role of genetic diversity and the role of interindividual variation in CNS radiation damage response mechanisms that may alter an individual's risks for behavior deficits and neuropathologies. Successful completion of this project will provide a mechanistic foundation for space-radiation-induced CNS risk that is based on quantitative molecular and cellular defects and their associations with measured neurocognitive and behavioral abnormalities. The project considers the time of onset and the persistence of effects, as well as inter-individual differences in responses. The project will employ space-relevant fluences of simulated HZE radiation, ensuring relevance for astronaut risk assessment. The results are expected to play a key role in helping to reduce the uncertainty associated with predicting the CNS risk for deep space exploration missions. This research is also designed to contribute to the identification of molecular targets for biological countermeasures to reduce CNS risks from space radiation.
Task Progress:	Ed. Note: Project has closed. Final reporting not received. See Bibliography for resulting publications since last report. (Ed., 6/19/25).
Bibliography Type:	Description: (Last Updated: 06/19/2025)